

AI INFRA SUMMIT HACKATHON 2026

DUET



Dual-arm Execution from Everyday Talk

Voice → Reason → Coordinate → Execute

Track: Bimanual VLA Manipulation with Multi-Modal Reasoning · Bonus: Best Use of Speechmatics

Easy to ask for. Hard to execute.

Natural-language robot control is simple to request and difficult to carry out safely across two manipulators.

- ▶ Instructions carry hidden preconditions

The cutlery starts inside a closed drawer — it must be opened first, exactly once.

- ▶ Neither arm can reach the whole table

Each place setting sits inside exactly one arm's reach envelope.

- ▶ The world is different every run

Placement, mass, friction, shape and lighting are randomized per seed.

Say it. Two arms do it.

Speechmatics voice → intent → deterministic planner → coordinated bimanual execution.

DOET
Dual-arm Execution from Everyday Talk — say a dinner-table instruction, watch two SO-101 arms reason, hand off, and carry it out.

WORKSPACE — PLAN VIEW
Dual SO-101 arms. Dashed rings are reach envelopes; the hole at each centre is real, set by the elbow stop.

SEED 1

PLANNER: DETERMINISTIC VOICE: IDLE SEEDS: 10 RANDOMIZED

VOICE — SPEECHMATICS REAL-TIME
Press LISTEN, or type a command below.

COMMAND
pick up the mug and place it on the right setting
Run

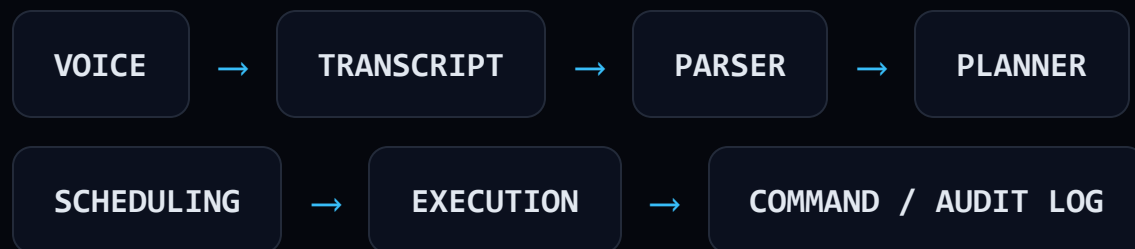
The brief's worked example Cross-workspace transfer Full place setting
Whole-task macro

Understood 1 instruction; planned 27 steps. 3 hand-offs required because the arms cannot both reach. Executing now.

PLAN — DEPENDENCY PHASES
Steps on one row have no ordering constraint between them, so both arms move together.

P00	Arm B moves above the top drawer handle
P01	Arm B opens the top drawer
P02	Arm B moves above the plate
P03	Arm B grasps the plate

Every stage is inspectable

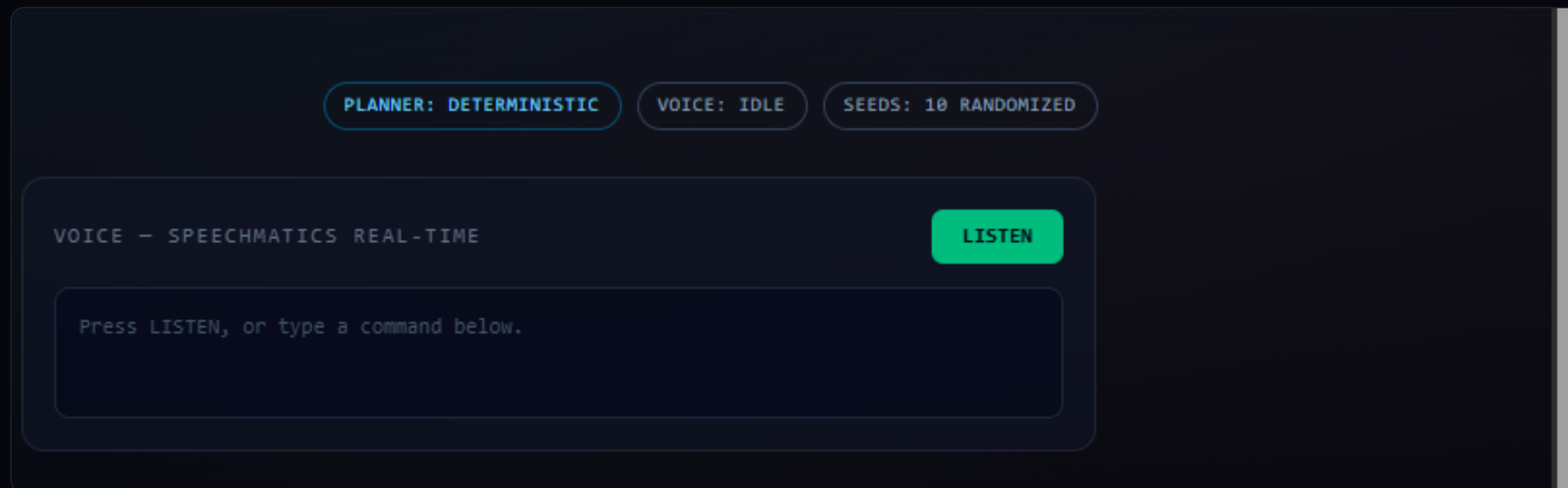


The planner is deterministic and **no language model sits in the action path**. A model may explain a plan; it never chooses one. The same sentence against the same seed produces a byte-identical plan, and the tests enforce it.

Reachability is decided by solving inverse kinematics — scene generator, planner and executor all call the same solver, so they cannot disagree.

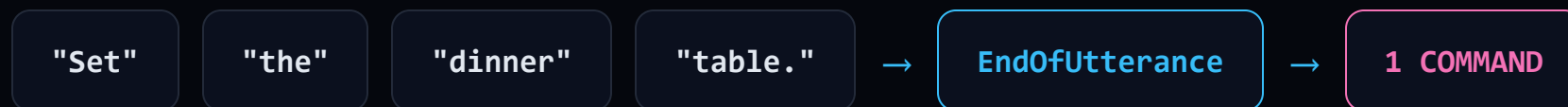
“Set the dinner table.” → 27 steps

One sentence expands into a dependency graph, not a list. Where an arm waits, the graph shows why.



One sentence = one command

AddTranscript is a *segment*, not a sentence. "Set the dinner table." arrives as four of them.



- ▶ Partial transcripts drive the live caption only — a half-heard phrase never moves an arm
- ▶ Verified live: 6 of 6 utterances, each producing exactly one command
- ▶ API key never reaches the browser; short-lived token, rate-limited to protect free-tier credits

Forced by geometry, not scripted

PLANNED STEPS

27

HAND-OFFS

3

PARALLELISM

2x

RESULT

EXECUTED

INTEL PHYSICAL AI · BIMANUAL VLA CHALLENGE

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PLANNER: DETERMINISTIC

VOICE: IDLE

SEEDS: 10 RANDOMIZED

VOICE — SPEECHMATICS REAL-TIME

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Measured, not asserted

SEEDED RUNS

39 / 40

SUCCESS RATE

98%

AUTOMATED TESTS

53 / 53

- ▶ Scoring is strict — the command must plan with no errors, **and** every step must execute, **and** the world must end in the requested state
- ▶ The one failing seed is kept deliberately; tuning it away would make the number meaningless
- ▶ 98% describes 40 seeded runs in DUET's deterministic simulator — not real-world robot accuracy

What DUET is, and is not

DELIBERATE SCOPE

- ▶ Purpose-built deterministic simulator
Runs in any browser, zero install
- ▶ Symbolic task planning
Reproducible and inspectable
- ▶ No physical robot claimed
- ▶ No Core Ultra / OpenVINO claimed

FOCUS INSTEAD

- ▶ Multimodal reasoning
- ▶ Real-time voice interaction
- ▶ Bimanual planning and hand-offs
- ▶ Robustness and reproducibility

OpenVINO / Core Ultra optimization was outside the available remote hardware environment. DUET therefore focuses on multimodal reasoning, real-time voice interaction, bimanual planning, robustness and reproducibility — and reports only what it measured.

Voice-first robotics you can verify

- ▶ **Voice-first** — one spoken sentence produces exactly one command
- ▶ **Explainable plans** — a dependency graph, not a black box
- ▶ **Coordinated hand-offs** — forced by reach geometry, never scripted
- ▶ **Deterministic execution** — same seed, same plan, byte for byte
- ▶ **Reproducible evaluation** — 39/40 seeded runs, 53/53 tests, committed evidence

Live duet-alpha-ebon.vercel.app

Code github.com/kmt9967/duet

Demo youtu.be/zi3mNkrSNaw